

Problem 2 (Spiral Triangle)

We show that for any pair of triangles $D_1E_1F_1$ and $D_2E_2F_2$ satisfying the conditions, the Miquel point of the two triangles is the same.

Draw (AF_1E_1) and (AF_2E_2) and let their second intersection be M . Note that there is a spiral similarity at M taking F_1E_1 to F_2E_2 . The same spiral similarity hence takes $D_1E_1F_1$ to $D_2E_2F_2$ since those triangles are directly similar. Since B, D_1, D_2 are collinear; B, F_1, F_2 are collinear, we know (BF_1D_1) and (BF_2D_2) intersect at M as well. Therefore, M is the Miquel point of both triangles.

Problem 5 (Z54090D0)

Let the line through B parallel to l_1 intersect AC at E ; let the line through C parallel to l_2 intersect AB at F . We claim X the point (APQ) always passes through is the second intersection of (ABE) and (ACF) .

For any point D , we know $\frac{BP}{PF} = \frac{BD}{DC} = \frac{EQ}{QC}$ by similar triangles. Therefore, the spiral similarity at X sending BF to EC sends PF to EQ as well. This suffices to show $(APXQ)$ is cyclic.

5 clubs; 6 hours

Problem 6 (79IMO3)

Let X_1, X_2 be the points on the two circles centered at O_1, O_2 respectively. We are given $\angle X_1O_1A = \angle X_2O_2A$. We claim the fixed point is P the reflection of A over the perpendicular bisector of O_1O_2 . Note that since AO_1O_2 is a trapezoid,

$$PO_1 = AO_2 = X_2O_2$$

$$PO_2 = AO_1 = X_1O_1$$

$$\angle PO_1X_1 = \angle AO_1X_1 - \angle AO_1Q = \angle AO_2X_2 - \angle AO_2P = \angle PO_2X_2$$

By SAS congruence, we get $PX_2 = PX_1$ as desired.

Problem 8 (Z7B5EE9D)

Define points as in the diagram on the right.

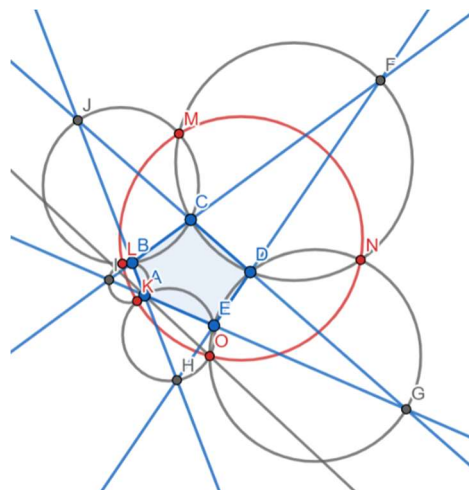
$$\angle MNO = \angle MKO$$

$$\Leftrightarrow \angle MCD + \angle DEO = \angle MKH + \angle HKO$$

From M, O being Miquel point,

$$\Leftrightarrow \angle MBH + \angle JAO = \angle MKH + \angle HAO$$

Now note that M, K both lie on (BHF) as M is the Miquel point of $BCDH$ and K that of $BAEF$.



$$\Leftrightarrow \angle MBH + \angle JAO = \angle MBH + \angle HAO$$

This is clearly true.

10 clubs; 7 hours

Problem 10 (13AMO1)

Let M be the Miquel Point.

WO check that there is a spiral similarity sending YX to BP by showing (PXM) is tangent to BC at P [clarification needed/is this a common thing to do?].

$$\angle MPC = \angle MQC = \angle MQA = \angle MXA = \angle MXP$$

Similarly, there is a spiral similarity sending XZ to PC at M . Since both spiral similarities send X to P , they are the same meaning $\frac{YX}{XZ} = \frac{BP}{PC}$ as desired.

Problem 12 (T6POP)

Let P be the second intersection of (BEC) and (AFC) . It suffices to show P lies on the C bisector. Indeed, there is a spiral similarity sending BF to EA at P . Since $BF = EA$, we know triangles BFP and EAP are congruent. The lengths of the altitude from P to BF and EA are congruent as desired.

15 clubs; 8 hours

Problem 14 (12TSTST7)

WLOG assume $\angle B \geq \angle C$.

We have, (HMN) tangent to $ML \Leftrightarrow \angle HNM = \angle HML$. Since $HLMD$ is tagent with $MK \perp DM$ and $HL \perp HD$,

$$\Leftrightarrow \angle HDL = \angle HNM \Leftrightarrow AD \parallel NM$$

Let (ADM) meet (ABC) again at L' . Note that,

$$\angle AL'M = \angle ADB = \angle C + \frac{\angle A}{2} = \angle AL'L$$

This means L' lies on ML so $L'M \perp BC$. As a result,

$$AD \parallel NM \Leftrightarrow \angle NML' = \angle \frac{B}{2} - \angle \frac{C}{2}$$

There is a spiral similarity sending BC to QP . Since M, N are midpoints, the same spiral similarity sends M to N . Therefore, there is a spiral similarity at L' sending MN to BQ . Indeed, $\angle LMN = \angle LBQ = \angle \frac{B}{2} - \angle \frac{C}{2}$ as desired.

Problem 15 (20USAEST2)

Let the line through B parallel to DE intersect AC at E_1 and define F_1 analogously. We claim the desired point Q is the second intersection of (AE_1B) and (AF_1C) .

We first check that this intersection Q always lies on (AEF) regardless of D .

$$\frac{BF}{FF_1} = \frac{BD}{DC} = \frac{E_1E}{EC}$$

There is a spiral similarity at Q sending BF_1 to E_1C . Therefore, the same spiral similarity sends F to E meaning there is a spiral similarity sending BF to E_1E . (AE_1B) and (AFE) intersect at Q as desired.

We finish showing $ABQ \sim APC$. Note that,

$$\angle AQB = \angle AE_1B = \angle ACP$$

Let $CF_1 \cap BE_1 = P'$. Note that $BP'CP$ is a parallelogram,

$$\frac{PC}{AC} = \frac{BP'}{AC}$$

There is a spiral similarity at Q sending BP' to AC since Q is the Miquel Point of $AE_1P'F_1$.

$$\frac{BP'}{AC} = \frac{BQ}{AQ}$$

By SAS similarity, we are done.

20 clubs; 9.5 hours

Problem 16 (10ELMO6)

Notice that MD intersects ω again at T since a homothety at T takes D to M (Death Star Lemma).

Let X' be where ME intersects ω again. It suffices to show that $TI, I_A X'$ intersects at S . Since TI intersects ω again at S , we show $TI \cap XI_A$ occurs on ω .

Continuation 1 (Spiral)

We show that this point is the A -antipode.

Let DPQ be the contact triangle. We claim that is the Miquel center of $BQPC$. Indeed, the Miquel Point T' lies on (ABC) again $T'D$ bisects $\angle BTC$ as $\frac{BT}{TC} = \frac{BQ}{CP} = \frac{BD}{CD}$. From $(AQPI)$ passes through T , $\angle ATI = \angle API = 90$. Therefore, TI passes through the A -antipode.

Similarly, $X'I_A$ passes through the A -antipode as X' is the Miquel Point of $BCP'Q'$.

Continuation 2 (Inversion)

Let X' be where ME intersects ω again. It suffices to show that $TI, I_A X'$ intersects at S . Since TI intersects ω again at S , we show $TI \cap XI_A$ occurs on ω . Equivalently, we show,

$$\angle MTI = \angle MXI_A$$

Note that $(BICI_A)$ is centered at M by the incentre-excentre lemma. We invert about this circle. Note that since,

$$\angle BTM = \frac{\angle A}{2} = \angle MBD$$

$$\Leftrightarrow MB^2 = MD \cdot MT$$

Therefore, $T^* = D$. Also, if we let M_A be the midpoint of BC , note that $MM_A \perp BC$ and MM_A is a diameter of ω . Therefore, D, E are reflections of each other over MM_A ; T, X are reflections of each other over MM_A . We then also have,

$$MD \cdot MT = ME \cdot MX$$

Therefore, $X^* = E$ also. Using the inversion,

$$\angle MTI = \angle MXI_A \Leftrightarrow \angle DIM = \angle EI_A M$$

Since I, M, I_A are collinear,

$$\Leftrightarrow DI \parallel EI_A$$

This is clearly true as both are perpendicular to BC .

Problem 19 (18AMO5)

Note that there is a spiral similarity at A sending BE to CQ and one sending PE to CD . Therefore, AC is the angle bisector of both $\angle BAQ$ and $\angle PAD$. Drawing the exterior angle bisector of these angles, we see that $\angle MAC = 90$ if and only if the exterior angle bisector passes through M .

Let the exterior angle bisector intersect BC and CD at X, Y respectively. We show that GQ, PF intersect XY at the same point to finish. Let M_1 be where GQ intersects XY . By Menelaus',

$$\frac{XM_1}{M_1Y} \cdot \frac{YQ}{QC} \cdot \frac{CG}{GX} = -1$$

Let M_2 be where PF intersects XY . By Menelaus',

$$\frac{XM_2}{M_2Y} \cdot \frac{YF}{FC} \cdot \frac{FX}{XP} = -1$$

Since $(C, Y; F, Q) = (C, P; B, G) = -1$, we have $\frac{YQ}{QC} = -\frac{YF}{FC}$ and $\frac{CG}{GX} = -\frac{FX}{XP}$ as desired.

Problem 20 (12TSTST2)

$$MN \parallel O_1O_2 \Leftrightarrow MN \perp PQ$$

Note that there is a spiral similarity at Q sending AC to BD . Since $AC = BD$, we have $AQ = BQ$ and $DQ = CQ$ as well. Therefore, Q is the arc midpoint of the arcs DC and AB with P . We therefore have PQ is the internal angle bisector of $\angle BPC$.

We see BM, CN, PQ are concurrent at the incenter of BPC . Since PQ is the radical axis, $(CMNB)$ is cyclic. We see that,

$$\angle BMN = \angle BCN = \frac{\angle BCA}{2}$$

Since $\angle CBM = \frac{\angle CBA}{2}$, we see that if $X = MN \cap BC$, $\angle MXC = \frac{\angle CBA - \angle BCA}{2}$. Similarly, if we let $Y = PQ \cap BC$, we have,

$$\angle AYC = 180 - \frac{\angle CAB}{2} - \angle CBA$$

Indeed, we see that $PQ \perp MN$ as desired.

Problem 21 (13TSTST4)

We claim the locus is a circle centered at Y . Let P be a point on Ω and let a line tangent from P to ω intersect ω at S and Ω again at P' . Let TS intersect Ω again at M . Let O, O' be the spiral centers of TSP and TSP' respectively.

Using the Salmon theorem on TSP' , there is a spiral similarity at T sending TSP' to TOY .

Using the Salmon theorem on TPS , we see that there's a spiral similarity at T sending TPS to TYO' .

However, by the Death Star lemma, there is a spiral similarity at T sending TPS to TMP' and TSP' to TPM since these triangles are directly similar.

Therefore, there is a spiral similarity at T sending $TOYO'$ to $TPMP'$. Therefore, we arrange the following equation from the Death Star Lemma,

$$PM^2 = MS \cdot MT$$

$$\Leftrightarrow OY^2 = YX \cdot YT$$

The RHS is constant, so we are done.